

- ① Find the intervals in which the function defined by $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$ is strictly increasing and strictly decreasing.
- ② Evaluate $\cos^{-1}[\cos(-\frac{7\pi}{3})]$
- ③ Let $f: \mathbb{R} - \{-4/3\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$. Show that f is one-one function. Also check whether f is an onto function or not.
- ④ Evaluate $\int \frac{x^2}{(x^2+4)(x^2+9)} dx$
- ⑤ If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$ then the value of α is.
- ⑥ If a line makes angles α , β and γ with three dimensional coordinate axes, respectively, then the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is.
- ⑦ Differentiate $\sin^{-1}\left(\frac{2^{x+1} \cdot 3^x}{1+(36)^x}\right)$
- ⑧ Evaluate $\int \frac{1}{\cos(x+a)\cos(x+b)} dx$
- ⑨ Find the general solution of the differential equation

$$-(1+\tan y)(dy-dx) + 2x dy = 0$$